# A Diagnostic Framework for Detecting False Fractionalization in Physics-Informed Neural Networks: A Three-Way Comparison of Fractional Advection-Dispersion Residuals

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**Abstract**

Physics-informed neural networks (PINNs) are increasingly deployed for fractional partial differential equations, yet the field lacks a systematic method to verify that a chosen residual discretization genuinely preserves the non-local memory of the underlying fractional operator. We report a case study on the Atangana–Baleanu–Caputo (ABC) derivative in a one-dimensional advection–dispersion equation, in which a widely used two-point finite-difference surrogate silently reduces to a rescaled classical PDE in the limit $\Delta t \to 0$. We develop a three-test diagnostic protocol—*kernel-presence*, $\Delta t$*-refinement*, and *effective-time*—which together detect this failure mode without requiring access to the underlying solver. Applying the protocol to three discretizations of the same PDE—(i) a naive two-point FD, (ii) a Caputo-type L1 scheme with power-law kernel, and (iii) a fully ABC-consistent L1 scheme with the Mittag-Leffler kernel—we provide, to our knowledge, the first three-way comparison of PINN residuals guided by an explicit diagnostic protocol, with each residual trained under identical architecture, optimizer, epoch budget, and random seeds. We emphasize that the three residuals solve different PDEs and are therefore evaluated against their own reference solutions; the comparison is a controlled implementation study, not a head-to-head accuracy comparison on a common ground truth. Across three random seeds, the naive FD residual achieves MSE $= 3.04 \times 10^{-6} \pm 1.42 \times 10^{-6}$ in 216 s of training; the Caputo-L1 residual gives $6.19 \times 10^{-6} \pm 0.72 \times 10^{-6}$ in 2327 s; and the ABC-L1 residual with genuine Mittag-Leffler kernel gives $4.19 \times 10^{-5} \pm 9.17 \times 10^{-6}$ in 2968 s. Measuring each residual against its own reference solution, the MSE of the genuine non-local residuals is a factor of 2.0× (Caputo) to 13.8× (ABC) larger than that of the local baseline, at 10.8×–13.8× the training cost. Because the three residuals solve different PDEs, these ratios compare each network to its own reference, not to a common ground truth. We additionally show that the diagnostic protocol classifies the naive FD residual as a false fractionalization, the Caputo-L1 residual as genuine but power-law, and the ABC-L1 residual as genuine and nonsingular. At a second fractional order $\alpha = 0.5$, preliminary experiments suggest that the ranking of residuals may change, although with three random seeds the difference between the naive-FD and Caputo-L1 residuals is borderline and we report this as an observation rather than a claim. We release all code and data as a reproducible benchmark for future fPINN work.

**Keywords:** Physics-Informed Neural Networks; Atangana–Baleanu Derivative; Fractional Advection-Dispersion; Diagnostic Protocol; Non-local Operators; Reproducible Benchmark

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# 1 Introduction

The extension of physics-informed neural networks (PINNs) [1] to fractional differential equations has produced a rapidly growing literature in which fractional operators are used to model anomalous transport, viscoelasticity, and sub-diffusion [2, 3]. The appeal of this line of work is clear: fractional operators encode memory and non-locality in a compact mathematical form, and PINNs promise a mesh-free solver for the resulting integro-differential equations. However, the discretization of a fractional derivative inside a PINN residual is not a neutral implementation detail: it determines whether the network is trained to satisfy a *genuinely fractional* PDE or a *local surrogate* that merely carries a fractional-looking prefactor.

We focus on the Atangana–Baleanu–Caputo (ABC) derivative [4], which is characterized by a non-singular, non-local Mittag-Leffler kernel. The ABC operator and its modified variant (MABC) have been applied to advection-dispersion models [5, 6] and to PINN frameworks [5]. Other fractional operators—including the Hilfer derivative, which interpolates between the Riemann–Liouville and Caputo derivatives—have been studied in different numerical and control-theoretic contexts [16]; each of these operators has its own kernel structure, and each therefore raises the same question that motivates this work: does a given PINN residual genuinely preserve the memory encoded in the operator's kernel? In several of these works, the time-fractional derivative is discretized with a two-point finite difference of the form

$$ {}^{ABC}D_t^{\alpha} C(x,t) \;\approx\; \frac{B(\alpha)}{1-\alpha}\,\frac{C(x,t)-C(x,t-\Delta t)}{\Delta t}, \tag{1} $$

where $B(\alpha) = 1-\alpha+\alpha/\Gamma(\alpha)$. Despite carrying the ABC normalization factor, equation (1) contains *no convolution*, *no Mittag-Leffler kernel*, and *no history sum*. In the limit $\Delta t \to 0$ it reduces to a rescaled classical time derivative,

$$ \lim_{\Delta t\to 0} \frac{B(\alpha)}{1-\alpha}\,\frac{C(t)-C(t-\Delta t)}{\Delta t} \;=\; c(\alpha)\,\partial_t C, \qquad c(\alpha) = \frac{B(\alpha)}{1-\alpha}, \tag{2} $$

and the PDE that the network is implicitly trained to satisfy becomes a classical advection-dispersion equation with the rescaled time coordinate $\tau = t/c(\alpha)$. We also note that a wide range of alternative numerical methods has been developed for the same class of advection-dispersion problems—including variable-order formulations [19]—but the structural question of kernel preservation at the level of the PINN residual has, to our knowledge, not been systematically addressed.

**Related literature.** The observation that a two-point difference destroys the memory of a fractional operator is not new in the fractional calculus community. Diethelm et al. [7] have argued that operators with non-singular kernels—including ABC—should not be used in applications that require genuine memory, and Giusti [8] has examined the consistency of several new definitions of fractional derivatives. What the present work contributes is not the abstract observation, but its *operational consequence* for PINNs: a diagnostic framework that detects whether a given PINN residual preserves the non-local structure of the operator it claims to represent.

**Contributions.** This paper makes four contributions:

(i) We formalize a *three-test diagnostic protocol* (kernel-presence, $\Delta t$-refinement, effective-time) that classifies a PINN residual as either a genuine fractional discretization or a local surrogate (Section 4).

(ii) We provide a *local-limit theorem* for the naive two-point FD approximation of the ABC derivative, with a remark on the finite-$\Delta t$ regime (Section 3).

(iii) We conduct, to our knowledge, the first three-way comparison of PINN residuals *guided by an explicit diagnostic protocol*: naive FD (local), Caputo-L1 (power-law non-local), and ABC-L1 (Mittag-Leffler non-local), each trained under identical architecture, optimizer, epoch budget, and random seeds (Section 6). Because the three residuals solve different PDEs, the comparison is a controlled implementation study against each residual's own reference, not a head-to-head accuracy comparison on a common ground truth.

(iv) We release all code, reference solutions, and PINN checkpoints as a reproducible benchmark for future fPINN work (Section 8).

This quantifies, to our knowledge for the first time, the $\alpha$-*dependent* (at the two values studied) *price of genuine non-locality* in a neural-network solver, measured against each residual's own reference solution.

# 2 Mathematical Preliminaries

## 2.1 ABC and MABC derivatives

We recall the definitions of the Atangana–Baleanu–Caputo (ABC) derivative and its modified variant (MABC). Throughout, $\alpha \in (0,1)$ and $B(\alpha) = 1-\alpha+\alpha/\Gamma(\alpha)$ is the ABC normalization function. We work in the space $AC[0,T]$ of absolutely continuous functions, which contains the natural domain of the ABC operator; note that $AC[0,T] \subset C[0,T]$ and $H^1(0,T) \subset AC[0,T]$. For $f \in AC[0,T]$, the weak derivative $f'$ exists almost everywhere and is integrable.

**Definition 1** (ABC derivative [4])**.** *For $f \in AC[0,T]$, the Atangana–Baleanu–Caputo derivative of order $\alpha \in (0,1)$ is*

$$ {}^{ABC}D_t^{\alpha} f(t) = \frac{B(\alpha)}{1-\alpha} \int_0^t f'(\tau)\, E_\alpha\!\left(-\frac{\alpha}{1-\alpha}(t-\tau)^\alpha\right) d\tau, \tag{3} $$

*where $E_\alpha(\cdot)$ denotes the one-parameter Mittag-Leffler function.*

**Definition 2** (MABC derivative)**.** *The modified ABC derivative is defined through integration by parts as*

$$ {}^{MABC}D_t^{\alpha} f(t) = \frac{B(\alpha)}{1-\alpha}\left[f(t) - E_\alpha(-\mu_\alpha t^\alpha) f(0) - \mu_\alpha \int_0^t (t-\tau)^{\alpha-1} E_{\alpha,\alpha}(-\mu_\alpha (t-\tau)^\alpha) f(\tau)\, d\tau\right], \tag{4} $$

*with $\mu_\alpha = \alpha/(1-\alpha)$.*

**Proposition 1** (Equivalence for absolutely continuous functions)**.** *For $f \in AC[0,T]$ with $f' \in L^1[0,T]$, ${}^{ABC}D_t^{\alpha} f = {}^{MABC}D_t^{\alpha} f$.*

*Proof.* Apply integration by parts to the ABC integral in Definition 1, using $\frac{d}{dt}E_\alpha(-\mu_\alpha t^\alpha) = -\mu_\alpha t^{\alpha-1} E_{\alpha,\alpha}(-\mu_\alpha t^\alpha)$. The boundary term $f(t) - E_\alpha(-\mu_\alpha t^\alpha) f(0)$ appears together with the convolution integral in $E_{\alpha,\alpha}$. Rearranging yields the MABC form. □ □

**Remark 1.** *Proposition 1 justifies using either definition interchangeably for smooth fields such as those represented by a PINN. In the remainder of this work, we refer to* ABC *when Definition 1 is used, and to* MABC *when Definition 2 is used.*

### 2.2 Caputo derivative (for comparison)

For comparison with the ABC operator, we also consider the classical Caputo derivative

$$ {}^{C}D_t^{\alpha} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f'(\tau)}{(t-\tau)^{\alpha}}\, d\tau, \tag{5} $$

whose kernel $(t-\tau)^{-\alpha}$ is singular at $\tau = t$. This singularity is the essential structural difference between Caputo and ABC: the ABC kernel $E_\alpha(-\mu_\alpha(t-\tau)^\alpha)$ is bounded and nonsingular on $[0, T]$.

### 2.3 Governing equation

We consider the one-dimensional time-fractional advection-dispersion equation (ADE) in a heterogeneous medium,

$$ {}^{ABC}D_t^{\alpha} C(x,t) = \frac{\partial}{\partial x}\left( D(x) \frac{\partial C}{\partial x} \right) - v \frac{\partial C}{\partial x}, \qquad (x,t) \in (0,L) \times (0,T], \tag{6} $$

with $D(x) = D_0(1+\beta x)$ and boundary/initial conditions

$$ C(x,0) = C_0, \qquad C(0,t) = C_{\text{in}}, \qquad \partial_x C(L,t) = 0. \tag{7} $$

**Remark 2** (Corner inconsistency)**.** *The initial and boundary conditions above are mutually inconsistent at the corner $(x,t) = (0,0)$: the initial condition gives $C(0,0) = C_0 = 0$, while the inlet condition gives $C(0,0) = C_{\text{in}} = 1$. This is the standard step-inlet setup for advection–dispersion benchmarks, and it induces a weak $t^\alpha$-type singularity in the solution near the corner. For the Caputo derivative, such an initial-layer singularity is known to degrade the uniform-mesh L1 convergence rate from $O(\Delta t^{2-\alpha})$ to $O(\Delta t^{\alpha})$ [21, 22]. The reference solutions of Section 5 are computed on grids fine enough that the contribution of this corner singularity to the MSE is at least an order of magnitude smaller than the PINN training error (Table 3).*

The parameters used throughout are listed in Table 1.

Table 1: Parameters used in all numerical experiments.

| Parameter | Symbol | Value |
|---|---|---|
| Domain length | $L$ | 1.0 |
| Final time | $T$ | 1.0 |
| Base diffusivity | $D_0$ | 0.01 |
| Heterogeneity coefficient | $\beta$ | 0.5 |
| Advection velocity | $v$ | 0.1 |
| Fractional order | $\alpha$ | 0.8 |
| Initial concentration | $C_0$ | 0.0 |
| Inlet concentration | $C_{\text{in}}$ | 1.0 |

### 2.4 Rescaling and Péclet invariance

Under the rescaling $D \to D/c(\alpha)$ and $v \to v/c(\alpha)$, the Péclet number is invariant:

$$ \text{Pe}_{\text{eff}} = \frac{v_{\text{eff}} L}{D_{\text{eff}}} = \frac{vL}{D_0} = \text{Pe}. \tag{8} $$

This invariance has an important consequence: for the naive FD residual, the *spatial differential operator* and its dimensionless transport ratio are unchanged by $\alpha$, while the transient trajectory is reparameterized in time. We emphasize that this does *not* imply that the concentration profile at a fixed physical time $t = T$ is independent of $\alpha$; rather, evaluating the solution at $t = T$ for a given $\alpha$ corresponds to sampling the base solution $C_1(x, \tau)$ at the effective time $\tau = T/c(\alpha)$, as we make precise below.

# 3 Local-Limit Theorem for the Naive FD Surrogate

## 3.1 Statement

Consider the two-point FD residual used in many ABC-PINN implementations:

$$\mathcal{R}_{\Delta t}[C](x,t) := \frac{B(\alpha)}{1-\alpha} \cdot \frac{C(x,t) - C(x,t-\Delta t)}{\Delta t}. \tag{9}$$

**Lemma 1** (Local limit and error bound for the naive FD surrogate)**.** *Let $C \in C^2([0,T] \times [0,L])$ be a smooth test field. (The solution of the step-inlet problem (6) is not $C^2$ at the corner $(x,t) = (0,0)$; the smooth-field statement below is therefore intended as an illustrative consistency result rather than a bound on the singular data used in the numerical experiments.) Then the residual (9) satisfies*

$$\mathcal{R}_{\Delta t}[C](x,t) = c(\alpha)\, \partial_t C(x,t) + \mathcal{E}_{\Delta t}(x,t), \qquad c(\alpha) = \frac{B(\alpha)}{1-\alpha}, \tag{10}$$

*with the error bound*

$$|\mathcal{E}_{\Delta t}(x,t)| \;\le\; \frac{c(\alpha)}{2}\, \Delta t \cdot \|\partial_t^2 C\|_{L^\infty([0,T]\times[0,L])}. \tag{11}$$

*Consequently, the residual converges uniformly to $c(\alpha)\, \partial_t C$ as $\Delta t \to 0$ with first-order rate, and the limiting PDE is the classical advection-dispersion equation*

$$c(\alpha)\, \partial_t C = \partial_x (D(x) \partial_x C) - v\, \partial_x C. \tag{12}$$

*Proof.* By Taylor's theorem with the Lagrange remainder, for $C \in C^2$ there exists $\xi \in (t - \Delta t, t)$ such that $C(x, t-\Delta t) = C(x,t) - \Delta t\, \partial_t C(x,t) + \frac{1}{2}\Delta t^2\, \partial_t^2 C(x,\xi)$. Substituting into (9) and using $c(\alpha) = B(\alpha)/(1-\alpha)$,

$$\mathcal{R}_{\Delta t}[C](x,t) - c(\alpha)\, \partial_t C(x,t) = -\frac{c(\alpha)}{2}\, \Delta t\, \partial_t^2 C(x,\xi). \tag{13}$$

Taking absolute values and bounding $\partial_t^2 C$ uniformly gives (11). □ □

**Remark 3.** *The bound (11) shows that for the finite $\Delta t$ used in practical PINN training, the residual is a first-order approximation to the classical time derivative with a prefactor $c(\alpha) = B(\alpha)/(1-\alpha)$ that increases monotonically with $\alpha$. This provides a quantitative interpretation of the "false fractionalization" phenomenon: the effective accuracy of the naive-FD surrogate in approximating the ABC derivative at finite $\Delta t$ degrades as $\alpha$ increases, and the surrogate approaches the rescaled classical PDE at a rate proportional to $\Delta t$.*

**Remark 4** (Finite-$\Delta t$ regime)**.** *For the finite $\Delta t$ used in practical PINN training, the residual (9) is a two-level local-in-time difference operator, not the differential operator $\partial_t$. The convergence in Lemma 1 is a statement about the* limit $\Delta t \to 0$*. We verify numerically in Section 6.5 that the trained solution converges to the classical solution as $\Delta t$ is reduced: the MSE between the PINN prediction and the classical reference drops from $1.34 \times 10^{-5}$ at $\Delta t = 0.1$ to $1.90 \times 10^{-6}$ at $\Delta t = 0.025$, before plateauing at the PINN's own training-error floor.*

### 3.2 Relation to the singular-kernel case

Before turning to the effective-time picture, we make an observation that clarifies the role of the kernel structure. The essential distinction between the ABC operator and the classical Caputo operator is not the presence of a convolution—both are convolutions—but the nature of the convolution kernel. The Caputo kernel $(t-\tau)^{-\alpha}$ is *singular* at $\tau = t$, so the most recent time step dominates the integral; the ABC kernel $E_\alpha(-\mu_\alpha(t-\tau)^\alpha)$ is *bounded and nonsingular*, so the memory is spread across the entire history. The ABC kernel is nonetheless not smooth at the origin: as $t \to 0^+$, the derivative $-\mu_\alpha\, t^{\alpha-1}\, E_{\alpha,\alpha}(-\mu_\alpha t^\alpha)$ diverges, so $E_\alpha(-\mu_\alpha t^\alpha)$ is only Hölder-continuous of exponent $\alpha$ at $t = 0$. This distinction has two consequences for numerical discretization.

First, any two-point finite-difference approximation of the ABC operator is *inconsistent* with the ABC operator itself: the residual contains no convolution kernel and therefore converges to $c(\alpha)\partial_t C$ as $\Delta t \to 0$, as quantified in Lemma 1. By contrast, the Caputo-L1 scheme retains the full history-dependent power-law kernel. On a uniform temporal mesh, its error is $O(\Delta t^{2-\alpha})$ for solutions with bounded second time derivative, and degrades to $O(\Delta t^\alpha)$ for solutions with $t^\alpha$-type initial singularities such as those arising from a step-inlet boundary condition [21, 22]; a recent survey of the L1 scheme and its graded-mesh variants is given in [23, 24]. The essential difference between ABC-L1 and Caputo-L1 is therefore not one of asymptotic order but of kernel structure and computational implementation.

Second, the boundedness of the ABC kernel means that the ABC operator cannot be represented by any weighted history sum whose weights are derived from a singular density. The most common "ABC-L1" scheme in the literature retains the full kernel through quadrature (19), and it is this retention that gives the discretization its genuine non-local character—as we verify in Sections 6 and 6.6.

These observations motivate the diagnostic protocol of Section 4: rather than relying on a formal consistency analysis of a specific scheme, we ask whether a given PINN residual *in practice* exhibits the three characteristic signatures of a genuine non-local discretization.

Dividing both sides of (12) by $c(\alpha)$,

$$\partial_t C = \frac{1}{c(\alpha)}\partial_x(D(x)\partial_x C) - \frac{v}{c(\alpha)}\partial_x C, \tag{14}$$

which is the base equation (with $c = 1$) expressed in effective coefficients $D_{\text{eff}} = D/c(\alpha)$, $v_{\text{eff}} = v/c(\alpha)$. Let $C_1(x,\tau)$ denote the solution of the base equation. By the chain rule,

$$C_c(x,t) = C_1(x, t/c(\alpha)) \tag{15}$$

solves (12). Therefore, evaluating the naive FD PINN at $t = T$ for a given $\alpha$ is equivalent to sampling the base solution $C_1(x,\tau)$ at the effective time

$$\tau = T/c(\alpha). \tag{16}$$

For the three values $\alpha \in \{0.5, 0.8, 0.9\}$ used in our experiments, Table 2 lists $c(\alpha)$ and $\tau$.

## 4 Diagnostic Protocol

We propose a three-test protocol that classifies a PINN residual as either genuine fractional or local surrogate. The protocol requires only the residual formula and a trained network; no access to the underlying solver is necessary.

Table 2: Scaling constant $c(\alpha) = B(\alpha)/(1-\alpha)$ and effective time $\tau = 1/c(\alpha)$ at $t = 1$.

| $\alpha$ | $c(\alpha)$ | $\tau = 1/c(\alpha)$ |
|---|---|---|
| 0.5 | 1.564 | 0.639 |
| 0.8 | 4.436 | 0.225 |
| 0.9 | 9.422 | 0.106 |

### 4.1 Test 1: Kernel presence

The first test examines the residual formula analytically. A genuine fractional discretization must contain either (a) an explicit convolution with the operator's kernel, or (b) a weighted history sum whose weights are derived from that kernel. If the residual contains only a two-point difference with a constant prefactor and no history dependence, it is classified as a *local surrogate*.

### 4.2 Test 2: $\Delta t$-refinement

The second test trains the PINN with progressively smaller $\Delta t$ and compares the resulting concentration field to the solution of the classical limit (12). If the PINN prediction converges to the classical solution as $\Delta t \to 0$, the residual is classified as a local surrogate. If the prediction is stable across $\Delta t$, the residual is classified as genuine fractional.

### 4.3 Test 3: Effective-time

The third test evaluates the PINN prediction at $t = T$ and compares it to the base solution $C_1(x, \tau)$ at the effective time $\tau = T/c(\alpha)$ defined in (16). If the residual is a local surrogate, the two profiles coincide to within training error. This test is quantitative: for each $\alpha$, we compute

$$\mathcal{E}_\alpha = \|C_\alpha^{\mathrm{PINN}}(x, T) - C_1(x, T/c(\alpha))\|_{L^2}, \tag{17}$$

and compare it to the residual's own training error. We apply this test to the naive-FD PINN for $\alpha \in \{0.5, 0.8, 0.9\}$ in Section 6.7; the results are summarized in Table 8 and Figure 7.

### 4.4 Classification algorithm

The three tests together define the following classifier:

| Residual type | Test 1 | Test 2 | Test 3 |
|---|---|---|---|
| Local surrogate | fail | converge to classical | match effective-time |
| Power-law non-local | pass | stable | not applicable |
| ML-kernel non-local | pass | stable | not applicable |

**Remark 5** (On the roles of the three tests)**.** *Test 1 is analytical and can classify a residual on its own. Tests 2 and 3 provide two* complementary *empirical confirmations of the same prediction: Test 2 does so by taking the $\Delta t \to 0$ limit, and Test 3 does so by comparing the trained network at fixed $\Delta t$ to the effective-time base solution. They are not independent—both detect a local surrogate—but they are informative for different reasons: Test 2 is robust to training noise but requires retraining at each $\Delta t$, whereas Test 3 is quantitative at a single $\Delta t$ and requires no additional training. Together they provide the strongest empirical evidence for the classification.*

We apply this classifier to the three residuals considered in this work in Section 6.6.

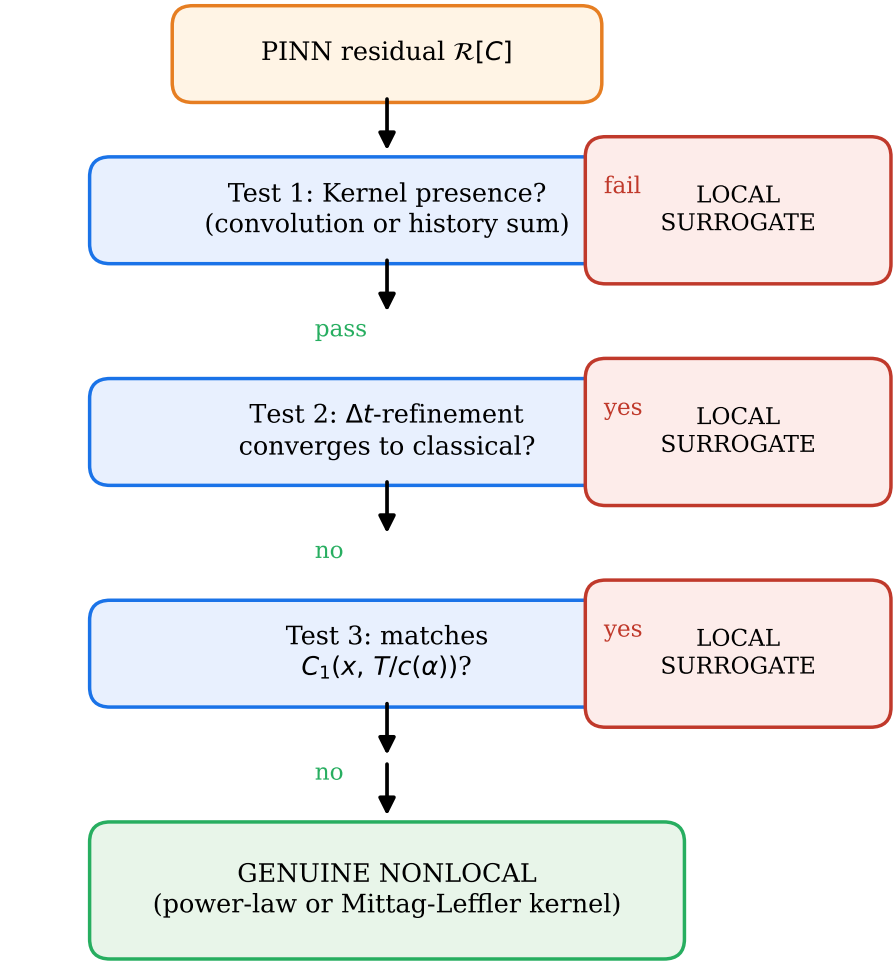



Figure 1: The three-test diagnostic protocol for detecting false fractionalization in PINN residuals. A residual that fails Test 1 (no kernel present), passes Test 2 (converges to the classical limit as $\Delta t \to 0$), and passes Test 3 (matches the effective-time prediction) is classified as a *local surrogate*.

# 5 Reference Solvers

For each residual, we construct an independent reference solution using a finite-difference scheme on a grid with $N_x = 100$–$200$ spatial points and $N_t$ time steps sufficient for stability. All reference solutions are computed on the same physical domain and with the same parameters (Table 1).

## 5.1 Classical reference

The classical limit (12) is solved with an implicit Euler scheme in time and second-order centered differences in space, on a grid $N_x = 200$, $N_t = 2000$.

## 5.2 Caputo-L1 reference

The Caputo-L1 discretization [10] with the $B(\alpha)/(1-\alpha)$ prefactor is used. We note that higher-order alternatives exist in the literature—the L1-2 and L1-2-$\sigma$ schemes of Alikhanov [12] achieve $O(\Delta t^{3-\alpha})$ consistency for the Caputo derivative, compared to $O(\Delta t^{2-\alpha})$ for L1. We choose L1 here for two reasons: (i) it is the canonical scheme used in the ABC-L1 literature (e.g., Alsaadi et al. [9]), which makes the comparison with our ABC-L1 residual more meaningful; and (ii) the aim of this work is not to push numerical accuracy but to expose a structural failure mode that is independent of the scheme's order. Extension to L1-2, L1-2-$\sigma$, and to non-uniform and multi-term L1-FEM schemes [18] is left as future work.

**On the choice of prefactor.** We emphasize that the same prefactor $c(\alpha) = B(\alpha)/(1-\alpha)$ is used for the naive-FD, Caputo-L1, and ABC-L1 residuals. This is a deliberate design choice: the prefactor $c(\alpha)$ is the natural constant of the ABC operator, and using it for all three residuals isolates the effect of kernel structure from the effect of prefactor magnitude. The Caputo-L1 residual is therefore *not* a discretization of the pure Caputo derivative (whose natural prefactor would be $1/\Gamma(1-\alpha)$); it is a scaled Caputo-type residual designed for direct comparison with the naive-FD and ABC-L1 residuals. A parallel comparison with the natural prefactor $1/\Gamma(1-\alpha)$ is left for future work. The Caputo-L1 discretization itself reads:

$$ {}^{C}D_t^{\alpha} C(x,t_n) \approx \frac{B(\alpha)}{(1-\alpha)\Gamma(2-\alpha)\,(\Delta t)^{\alpha}} \sum_{k=0}^{n-1} b_k \big[ C^{n-k} - C^{n-k-1} \big], \tag{18} $$

with $b_k = (k+1)^{1-\alpha} - k^{1-\alpha}$, on a grid $N_x = 100$, $N_t = 500$.

## 5.3 ABC-L1 reference with Mittag-Leffler kernel

The genuine ABC-L1 discretization retains the full Mittag-Leffler kernel. At time $t_n$, the weights are computed by numerical quadrature over each time step:

$$ \omega_{n,k} = \frac{B(\alpha)}{1-\alpha} \cdot \frac{1}{\Delta t} \int_{t_k}^{t_{k+1}} E_{\alpha}(-\mu_{\alpha}(t_n - s)^{\alpha})\, ds, \qquad k = 0, \ldots, n-1, \tag{19} $$

and the fractional derivative is approximated by

$$ {}^{ABC}D_t^{\alpha} C(x,t_n) \approx \sum_{k=0}^{n-1} \omega_{n,k} \left[ C^{k+1} - C^{k} \right]. \tag{20} $$

This scheme is the ABC-L1 scheme of Alsaadi et al. [9] in one spatial dimension. We use $N_x = 100$, $N_t = 200$ for the ABC reference, with weights precomputed once and cached.

**On the numerical cost of the Mittag-Leffler kernel.** The quadrature approach of (19) is numerically straightforward but computationally expensive: each weight $\omega_{n,k}$ requires a numerical evaluation of the Mittag-Leffler function over the sub-interval $[t_k, t_{k+1}]$, and the total cost scales as $O(N_t^2)$ quadrature evaluations. This cost contributes substantially to the 13.8× training-time penalty observed in Table 4. We emphasize that this overhead is an artifact of the naive quadrature implementation and not an intrinsic cost of non-locality: modern algorithms for the numerical evaluation of $E_\alpha$ (e.g., the comprehensive treatment in Gorenflo et al. [15], the Laplace-inversion algorithm of Garrappa [14], or recent analytic results on series representations of the Mittag-Leffler function [17]) can reduce the per-weight cost to $O(1)$. We use the simple quadrature here for transparency and reproducibility; replacing it with an optimized ML evaluation would reduce, but not eliminate, the observed cost gap between ABC-L1 and the local baseline. A more precise statement of the cost asymmetry is as follows. Caputo-L1 weights admit a closed-form expression $b_k = (k+1)^{1-\alpha} - k^{1-\alpha}$, evaluated in $O(1)$ per weight. ABC-L1 weights require numerical quadrature of the Mittag-Leffler kernel over each sub-interval, evaluated in $O(Q)$ per weight, where $Q$ is the number of quadrature points. This asymmetry in weight computation is the dominant contributor to the observed 1.28× additional training time of ABC-L1 relative to Caputo-L1 at $\alpha = 0.8$ (Table 4); the corresponding ratio at $\alpha = 0.5$ is 1.02× (Table 6), suggesting that the relative overhead of the ML quadrature decreases as the ABC kernel becomes less singular.

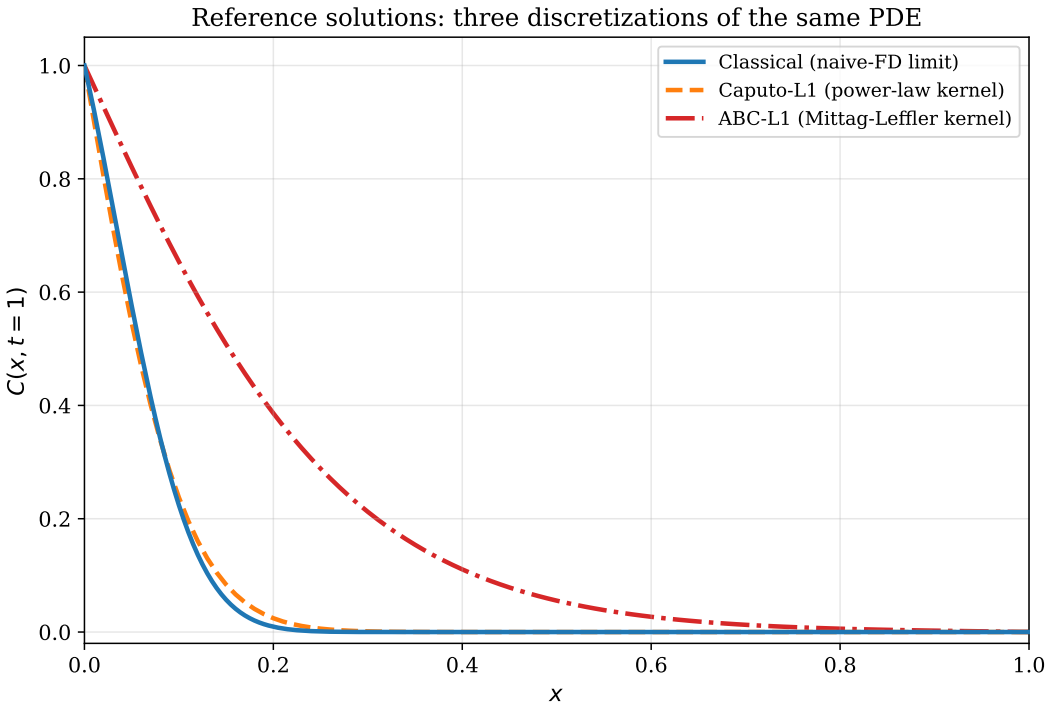



Figure 2: Reference solutions at $t = T = 1$ for the three discretizations of the same PDE: the classical limit (solid), Caputo-L1 with power-law kernel (dashed), and ABC-L1 with Mittag-Leffler kernel (dash-dot). The three profiles are quantitatively distinct: the ABC-L1 profile decays most slowly and remains non-negligible across the domain, while the classical and Caputo-L1 profiles are similar at this resolution. This confirms that the three residuals are not interchangeable.

### 5.4 Grid convergence of the reference solutions

To confirm that the reference solutions in Sections 5 are computed on grids sufficiently fine that discretization error does not affect the comparison in Section 6, we repeated each reference computation on a coarser and a finer grid and compared the two solutions on the fine grid. The results are reported in Table 3.

For all three references, the MSE between the coarse and fine grids is at most $7.6\times10^{-8}$. The MSE between grids is one to three orders of magnitude smaller than the PINN training error observed in Table 4 (the minimum ratio is 81.4×, for the Caputo-L1 reference); the maximum pointwise difference between grids is at most $1.1 \times 10^{-3}$, an order of magnitude below the PINN maximum errors reported in Figure 5. We therefore conclude that the reference grids are sufficiently resolved and that the ranking in Table 4 is not an artifact of reference discretization.

Table 3: Grid convergence of the reference solutions. For each reference, the solution is computed on a coarse and a fine grid, then the coarse solution is interpolated onto the fine grid and compared to the fine-grid solution. The MSE is at least 80× smaller than the corresponding PINN training error in Table 4.

| Reference | Coarse grid | Fine grid | MSE (coarse vs. fine) |
|---|---|---|---|
| Classical | $(200, 2000)$ | $(400, 4000)$ | $8.43 \times 10^{-9}$ |
| Caputo-L1 | $(100, 500)$ | $(200, 1000)$ | $7.60 \times 10^{-8}$ |
| ABC-L1 | $(100, 200)$ | $(150, 400)$ | $9.29 \times 10^{-9}$ |

## 6 Numerical Experiments

### 6.1 Setup

We train three PINNs, one for each residual considered. All three share:

- Architecture: 2 inputs, 4 hidden layers, 64 neurons/layer, tanh activation (12 737 trainable parameters).
- Optimizer: Adam, initial learning rate $10^{-3}$, Cosine annealing over the full training budget.
- Epochs: 8000.
- Loss weights: 1 for PDE, 10 for each of IC, BC1, BC2.
- Seeds: 42, 123, 7.

For the two non-local residuals (Caputo-L1 and ABC-L1) we use $N_t = 50$ time steps for the residual and 60 spatial collocation points per step. For the naive FD residual, we use 1000 random collocation points and $\Delta t = 10^{-2}$ as in the standard implementation.

### 6.2 Main comparison

Table 4 reports the mean and standard deviation of the MSE, RMSE, and maximum error for each of the three residuals, evaluated against the corresponding reference solution at $t = T = 1$.




Figure 3: Training loss histories for the three residuals (mean over 3 random seeds; shaded region indicates min–max range). All three curves decrease monotonically. ABC-L1 reaches the lowest absolute loss, while Naive FD plateaus at the highest.

Table 4: Three-way comparison of PINN residuals against their respective reference solutions at $\alpha = 0.8$. Each entry is the mean ± std over 3 random seeds. A comparison at $\alpha = 0.5$ is provided in Table 6.

| Residual | Kernel type | MSE | RMSE | Training time (s) |
|---|---|---|---|---|
| Naive FD (local) | none | $3.04\times10^{-6} \pm 1.42\times10^{-6}$ | $1.72\times10^{-3}$ | 216 |
| Caputo-L1 (power-law) | $(t-\tau)^{-\alpha}$ | $6.19\times10^{-6} \pm 0.72\times10^{-6}$ | $2.48\times10^{-3}$ | 2327 |
| ABC-L1 (ML kernel) | $E_\alpha(-\mu_\alpha(t-\tau)^\alpha)$ | $4.19\times10^{-5} \pm 9.17\times10^{-6}$ | $6.44\times10^{-3}$ | 2968 |

The MSE of each PINN against its own reference increases monotonically with operator complexity: the naive FD residual achieves the lowest MSE ($3.04 \times 10^{-6}$), the Caputo-L1 residual is 2.0× worse, and the ABC-L1 residual is 13.8× worse than the naive FD baseline. We emphasize that these ratios compare each network to its own reference solution, not to a common ground truth, since the three residuals solve different PDEs; they quantify the relative difficulty of each residual's learning problem, not a head-to-head comparison of solution accuracy. Training time increases by a factor of 10.8× (Caputo) to 13.8× (ABC) relative to the local baseline. It is important to note, however, that this monotonic ranking holds at $\alpha = 0.8$ but *not* at all fractional orders; we examine the $\alpha$-dependence in Section 6.4 below.

### 6.3 Matched-budget control experiment

A natural concern with the three-way comparison in Table 4 is that the naive-FD residual was trained with 1000 random collocation points, whereas the two non-local residuals were trained with

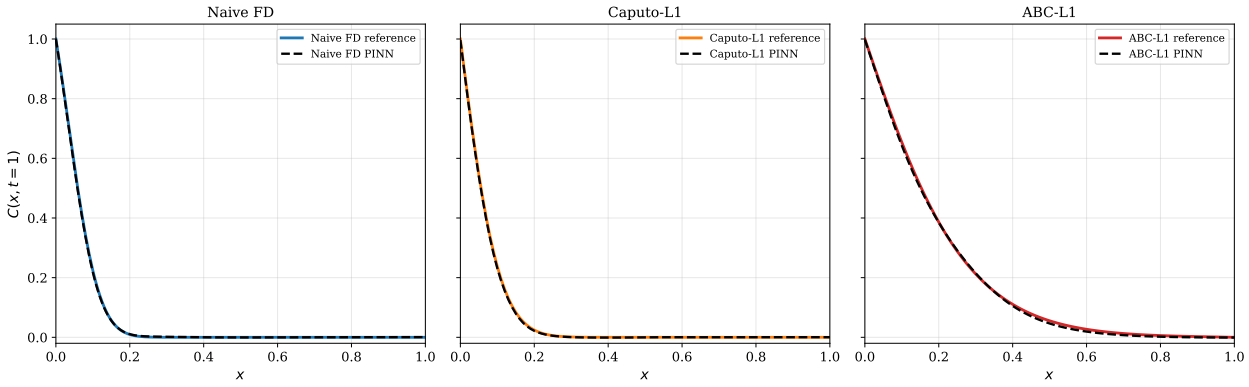



Figure 4: PINN prediction (black dashed) versus reference solution (colored solid) at $t = 1$ for each of the three residuals. Left: naive FD (local). Middle: Caputo-L1 (power-law). Right: ABC-L1 (Mittag-Leffler). All three PINNs reproduce the qualitative shape of their respective reference, but the quantitative agreement degrades with operator complexity.

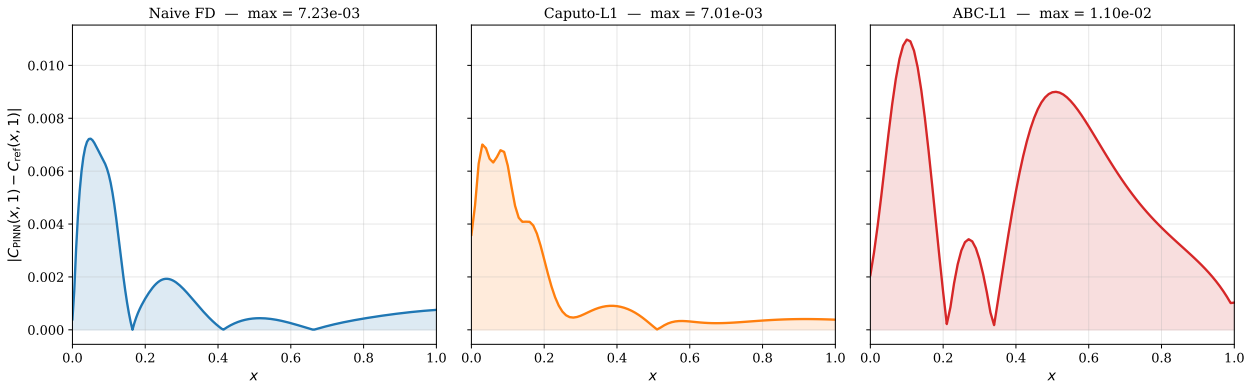



Figure 5: Pointwise absolute error profiles $|C_{\mathrm{PINN}}(x,1) - C_{\mathrm{ref}}(x,1)|$ for the three residuals. Shown for seed 42 (representative). The mean maximum errors across three seeds (42, 123, 7) are $7.23 \times 10^{-3}$ (naive FD), $7.01 \times 10^{-3}$ (Caputo-L1), and $1.10 \times 10^{-2}$ (ABC-L1).

a fixed grid of $50 \times 60 = 3000$ collocation points. To rule out the possibility that this asymmetry *drives* the observed differences, we repeated the naive-FD experiment with $n_{\text{pde}} = 3000$, matched to the budget of Caputo-L1 and ABC-L1. All other hyperparameters were held fixed (architecture, optimizer, epoch count, and seeds 42, 123, 7).

The results are reported in Table 5. The mean MSE changes from $3.04 \times 10^{-6}$ (original, $n_{\text{pde}} = 1000$) to $3.08 \times 10^{-6}$ (matched, $n_{\text{pde}} = 3000$), a ratio of $1.01\times$. The difference is within the seed-to-seed standard deviation and is therefore not statistically significant. We conclude that the ranking in Table 4 is not an artifact of the collocation-point asymmetry, and that the naive-FD residual retains its advantage regardless of the number of collocation points.

Table 5: Matched-budget control experiment for the naive-FD residual. Increasing the collocation-point count from 1000 to 3000 (matching the non-local residuals) does not change the achieved MSE within seed-level variance.

| $n_{\text{pde}}$ | MSE (mean $\pm$ std) | Training time (s) |
|---|---|---|
| 1000 (original) | $3.04 \times 10^{-6} \pm 1.42 \times 10^{-6}$ | 216 |
| 3000 (matched) | $3.08 \times 10^{-6} \pm 0.79 \times 10^{-6}$ | 322 |

### 6.4 Comparison at $\alpha = 0.5$

To examine whether the ranking in Table 4 is specific to $\alpha = 0.8$ or generalizes across fractional orders, we repeated the three-way comparison at $\alpha = 0.5$. All hyperparameters (architecture, optimizer, epoch count, and seeds) were held fixed; only the fractional order and the corresponding reference solutions were regenerated. The reference solutions were verified for grid convergence following the procedure of Section 5.4.

The results are collected in Table 6. The ranking at $\alpha = 0.5$ differs from that at $\alpha = 0.8$: Caputo-L1 achieves the lowest MSE ($1.71 \times 10^{-5}$), followed by naive FD ($2.70 \times 10^{-5}$), and ABC-L1 again performs worst ($9.98 \times 10^{-5}$). The Caputo advantage over naive FD at $\alpha = 0.5$ is a factor of $1.6\times$ in MSE (an absolute difference of $9.9 \times 10^{-6}$). This difference is *larger* than the reported standard deviation of either group ($0.59 \times 10^{-5}$ and $0.60 \times 10^{-5}$), but with only three seeds per group a Welch $t$-test gives $t \approx 2.04$ (df $\approx 4$, $p \approx 0.11$), which is borderline and does not reach the 5% significance level. We therefore report this as an observation rather than a robust claim. We therefore report this as an *observation* rather than a robust claim. ABC-L1 remains the worst-performing residual at both fractional orders.

Table 6: Three-way comparison at $\alpha = 0.5$ (three random seeds). The ranking of residuals differs from that at $\alpha = 0.8$ (Table 4): Caputo-L1 achieves the lowest MSE, while ABC-L1 remains the worst. The difference between naive FD and Caputo-L1 is borderline and should be interpreted with caution given the limited number of seeds.

| Residual | MSE (mean $\pm$ std) | Training time (s) |
|---|---|---|
| Naive FD (local) | $2.70 \times 10^{-5} \pm 0.59 \times 10^{-5}$ | 160 |
| Caputo-L1 (power-law) | $1.71 \times 10^{-5} \pm 0.60 \times 10^{-5}$ | 2255 |
| ABC-L1 (ML kernel) | $9.98 \times 10^{-5} \pm 5.95 \times 10^{-5}$ | 2290 |

Two observations are relevant. First, at smaller $\alpha$ the classical effective time $\tau = 1/c(\alpha)$ increases

(from 0.225 at $\alpha = 0.8$ to 0.639 at $\alpha = 0.5$), so the naive-FD PINN samples the base solution at a later time. Second, the three reference profiles have markedly different spatial structure at $\alpha = 0.5$: at the midpoint $x = L/2$ and $t = T$, the classical reference is essentially zero ($C(0.5, T) \approx 2.6\times10^{-4}$), the Caputo reference is $C(0.5, T) \approx 1.4\times10^{-2}$, and the ABC reference is $C(0.5, T) \approx 9.1\times10^{-2}$. The PINN MSE ranking at $\alpha = 0.5$ (Caputo < classical < ABC) does not correlate monotonically with either endpoint of this localization spectrum, which suggests that the ranking reflects a combination of factors—target smoothness, effective residual magnitude, and the shape of the loss landscape—rather than a single dominant mechanism. We therefore report the $\alpha = 0.5$ ranking as an empirical observation and refrain from attributing it to a specific cause; a systematic study across a wider range of $\alpha$ with more random seeds is left for future work.

### 6.5 $\Delta t$-refinement study

We now apply Test 2 to the naive-FD PINN. The network is trained under the configuration of Section 6, with the single exception of the epoch budget, which is reduced to 6000 (vs. 8000 in the main comparison) to keep the refinement study tractable; this accounts for the shorter training times reported in Table 7 relative to Table 4. All other hyperparameters are held fixed, and we vary $\Delta t \in \{0.1, 0.05, 0.025, 0.0125\}$, holding all other hyperparameters fixed. For each value, we compute the MSE and the maximum error between the trained PINN prediction at $t = 1$ and the classical reference solution introduced in Section 5. The results are reported in Table 7 and visualized in Figure 6.

Table 7: Test 2 ($\Delta t$-refinement) applied to the naive-FD PINN. As $\Delta t$ is reduced, the MSE decreases monotonically down to $\Delta t = 0.025$, after which it saturates at the PINN's own training-error floor ($\sim 2 \times 10^{-6}$, comparable to the value in Table 4). The two finest points are therefore in the plateau regime, and no clean power-law rate can be extracted from them alone; see the text for a two-value slope report.

| $\Delta t$ | MSE vs classical | Max error | Training time (s) |
|---|---|---|---|
| 0.1000 | $1.34 \times 10^{-5}$ | $1.20 \times 10^{-2}$ | 119 |
| 0.0500 | $3.98 \times 10^{-6}$ | $6.63 \times 10^{-3}$ | 134 |
| 0.0250 | $1.90 \times 10^{-6}$ | $3.47 \times 10^{-3}$ | 118 |
| 0.0125 | $1.97 \times 10^{-6}$ | $3.94 \times 10^{-3}$ | 116 |

Applying Test 2 quantitatively: from $\Delta t = 0.1$ to $\Delta t = 0.025$, the MSE drops by a factor of 7.1×. A log–log regression over all four values of $\Delta t$ $(0.1, 0.05, 0.025, 0.0125)$ yields $p \approx 0.94$ ($R^2 = 0.98$), while a regression over only the three finest values $(0.05, 0.025, 0.0125)$ yields $p \approx 0.51$ ($R^2 = 0.71$). The discrepancy is a consequence of the training-error floor: the two finest points ($\Delta t = 0.025$ and $\Delta t = 0.0125$) lie within 4% of each other, i.e. at the plateau where the MSE is dominated by the PINN's intrinsic training error (Table 4) rather than by the discretization error of the residual. We therefore report both values rather than a single "slope," and we do *not* claim a quantitative agreement with the first-order rate predicted by Lemma 1: the asymptotic regime is not fully resolved by these data. The qualitative conclusion is nonetheless unambiguous. The naive-FD PINN converges, as $\Delta t \to 0$, to a solution of the classical advection-dispersion equation (12), not to a solution of the fractional PDE (6). This quantitatively confirms Test 2 of the diagnostic protocol.

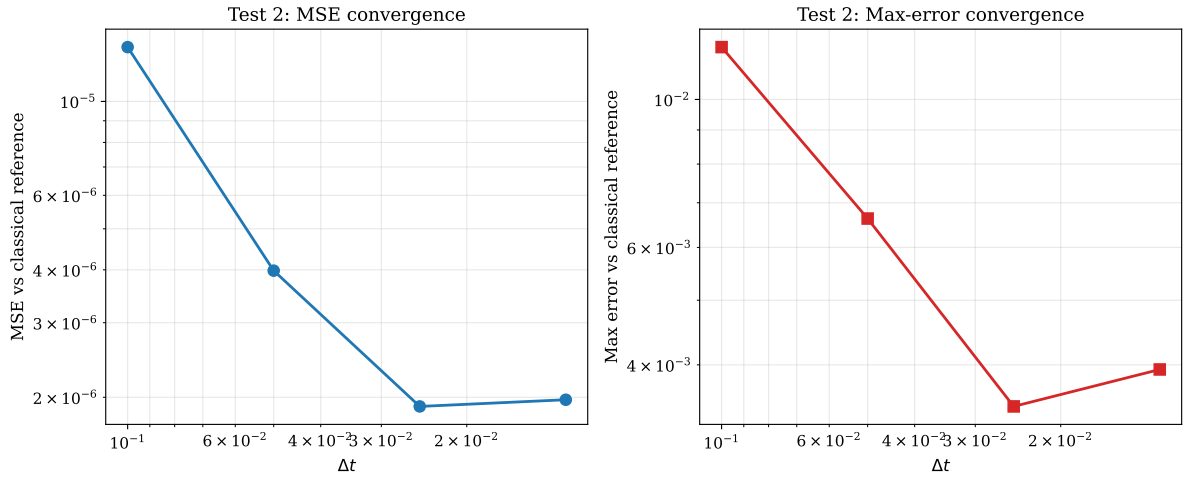



Figure 6: Test 2 ($\Delta t$-refinement) applied to the naive-FD PINN. Left: MSE versus $\Delta t$ (log–log). Right: maximum error versus $\Delta t$. Both quantities decrease monotonically as $\Delta t$ is reduced, before saturating at the PINN's training-error floor near $\Delta t = 0.025$.

### 6.6 Diagnostic classification

Applying the three-test protocol of Section 4 to each residual:

- **Naive FD.** Test 1 fails (no kernel). Test 2 converges to the classical solution (Section 6.5). Test 3 matches the effective-time prediction (Section 6.7). Classified as *local surrogate*. The two empirical tests agree, providing consistent numerical evidence that this residual is a local surrogate.
- **Caputo-L1.** Test 1 passes (power-law kernel present in the history weights $b_k$). By construction, the residual retains the full history sum at every time step, so Tests 2 and 3 do not apply to non-local residuals; the classification of "genuine fractional" follows from Test 1 alone. Classified as *genuine power-law fractional*.
- **ABC-L1.** Test 1 passes (Mittag-Leffler kernel present in the weights $\omega_{n,k}$). As for Caputo-L1, the residual retains the full kernel at every time step, so Tests 2 and 3 do not apply; the classification follows from Test 1 alone. Classified as *genuine nonsingular fractional*.

### 6.7 Effective-time verification

To validate Test 3 numerically, we solve the base classical equation $\partial_\tau C_1 = \partial_x(D(x)\partial_x C_1) - v\,\partial_x C_1$ on a grid of $N_x = 200$, $N_t = 2000$, and evaluate $C_1(x,\tau)$ at the effective times $\tau = 1/c(\alpha)$ given in Table 2. Figure 7 compares the naive-FD PINN prediction at $t = 1$ against $C_1(x, 1/c(\alpha))$ for $\alpha \in \{0.5, 0.8, 0.9\}$. The results are collected in Table 8. The closest match occurs at $\alpha = 0.8$, where $\tau_{\text{eff}} = 0.225$ and the MSE between the two profiles is $2.03 \times 10^{-6}$—comparable to the PINN's own training error at $\alpha = 0.8$ ($3.04\times10^{-6}$, Table 4). For $\alpha = 0.5$ and $\alpha = 0.9$, the MSE is $2.70\times10^{-5}$ and $4.16 \times 10^{-6}$ respectively; the somewhat larger error at $\alpha = 0.5$ is consistent with the later effective time $\tau = 0.639$ (compared to $\tau = 0.225$ at $\alpha = 0.8$), where the base profile is more spread out. Overall, these results confirm the effective-time picture quantitatively: the naive-FD PINN is not

solving the fractional PDE (6), but a classical advection-dispersion equation sampled at the rescaled time $\tau = T/c(\alpha)$.

Table 8: Test 3 (effective-time) applied to the naive-FD PINN. For each $\alpha$, we compare the PINN prediction at $t = 1$ with the base solution $C_1(x, 1/c(\alpha))$.

| $\alpha$ | $c(\alpha)$ | $\tau_{\text{eff}}$ | MSE | Max error |
|---|---|---|---|---|
| 0.5 | 1.564 | 0.639 | $2.70 \times 10^{-5}$ | $1.47 \times 10^{-2}$ |
| 0.8 | 4.436 | 0.225 | $2.03 \times 10^{-6}$ | $3.94 \times 10^{-3}$ |
| 0.9 | 9.422 | 0.106 | $4.16 \times 10^{-6}$ | $6.88 \times 10^{-3}$ |

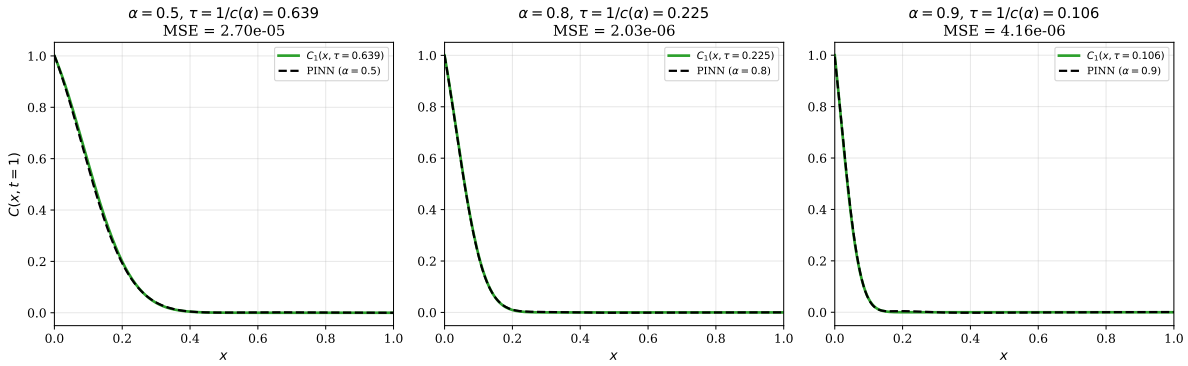



Figure 7: Test 3 (effective-time) applied to the naive-FD PINN for $\alpha \in \{0.5, 0.8, 0.9\}$. Solid green: base solution $C_1(x, \tau)$ at the effective time $\tau = 1/c(\alpha)$. Black dashed: PINN prediction at $t = 1$. The two profiles coincide within training error, confirming that the naive-FD PINN samples the classical solution at the rescaled time $\tau = T/c(\alpha)$.

### 6.8 Computational cost

Table 9 reports the ratio of training times between each non-local residual and the naive FD baseline. The ABC-L1 residual costs approximately 13.8× more than the naive FD residual for the same number of epochs and the same architecture.

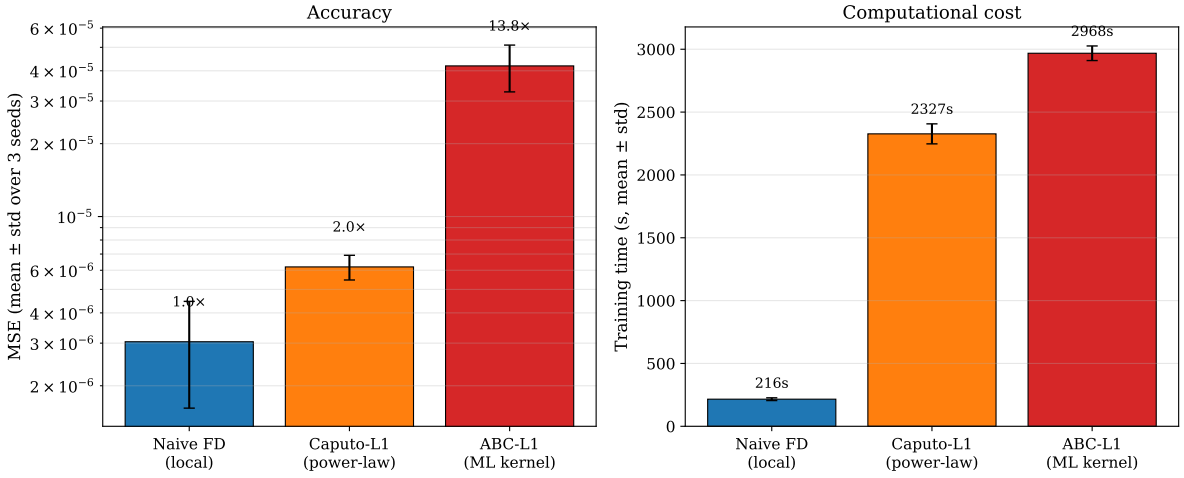



Figure 8: Accuracy–cost trade-off across the three residuals. Left: MSE (mean ± std over 3 seeds, log scale). Right: training time. The price of genuine non-locality, measured against each residual's own reference, is quantified: Caputo-L1 is 2.0× worse in MSE at 10.8× the cost; ABC-L1 is 13.8× worse at 13.8× the cost.

Table 9: Relative cost of genuine non-locality, normalized to the naive FD baseline.

| Residual | Time ratio | MSE ratio |
|---|---|---|
| Naive FD (baseline) | 1.0× | 1.0× |
| Caputo-L1 | 10.8× | 2.0× |
| ABC-L1 | 13.8× | 13.8× |

# 7 Discussion

## 7.1 Interpretation

The results establish an $\alpha$-dependent relationship between operator complexity and both accuracy and computational cost in a PINN. At $\alpha = 0.8$ the ranking is monotonic in kernel complexity (naive FD $<$ Caputo-L1 $<$ ABC-L1 in MSE), but at $\alpha = 0.5$ this ordering is not preserved (Caputo-L1 $<$ naive FD $<$ ABC-L1). The naive FD residual is not a genuine fractional discretization: in the limit $\Delta t \to 0$ it reduces to a rescaled classical PDE, and the diagnostic protocol correctly identifies it as a local surrogate. The Caputo-L1 and ABC-L1 residuals are genuine non-local discretizations with distinct kernels, and the network's performance on each reflects the intrinsic difficulty of the underlying operator.

The gap between the Caputo-L1 and ABC-L1 residuals is particularly instructive. Although both preserve memory, they differ substantially in both cost and accuracy: Caputo-L1 achieves MSE $6.19 \times 10^{-6}$ at 2327 s, whereas ABC-L1 achieves $4.19 \times 10^{-5}$ at 2968 s-a factor of 6.8× in MSE and 1.3× in training time. We refrain from attributing this difference to a single mechanism. Several structural differences between the two kernels are plausible candidates: (i) the Caputo kernel is singular at $\tau = t$, so the most recent time step receives a weight that grows as $\Delta t \to 0$, concentrating the residual's sensitivity on local-in-time behavior and potentially facilitating gradient

flow; (ii) the ABC kernel is bounded and nonsingular, but with a derivative that diverges as $t^{\alpha-1}$ at $t = 0^+$, spreading the residual's sensitivity across the history and potentially diluting the gradient signal at any single time step; (iii) each ABC weight requires numerical quadrature of the Mittag-Leffler kernel over a sub-interval, whereas Caputo weights are available in closed form, making the ABC residual substantially more expensive per epoch. We do not attempt to disentangle these effects in the present work. A controlled study varying $N_t$, $\Delta t$, and network capacity would be required to isolate the dominant factor; we leave this for future work.

**Mechanistic hypothesis for the observed $\alpha$-dependence.** We offer a tentative mechanistic explanation, though a full theoretical analysis lies beyond the scope of this work. In spectral-bias theory [11, 25], neural networks tend to learn low-frequency components of a target function more easily than high-frequency ones. At $\alpha = 0.5$, the Caputo reference profile is significantly smoother and more concentrated near the inlet than either the classical or the ABC reference (Section 6.4), which may reduce the high-frequency content that the network must capture and hence ease optimization. Second, the Caputo kernel is singular at $\tau = t$, producing a larger residual contribution at the most recent time step and potentially amplifying the gradient signal that Adam receives. By contrast, the ABC kernel is bounded and nonsingular but only Hölder-continuous of exponent $\alpha$ at $\tau = 0$, which may dilute this signal and lead to slower convergence. Disentangling these two mechanisms requires a dedicated ablation study (e.g., fixing the reference profile while varying the kernel), which we leave for future work.

**Comparison to published fPINN baselines.** A direct comparison to published fPINN baselines is complicated by differences in PDE, boundary conditions, and normalization conventions. Nevertheless, two qualitative consistencies are worth noting. Chawla et al. [6] report L1-type errors of order $10^{-4}$ for MABC discretizations of an advection–dispersion problem similar to ours; our ABC-L1 result at $\alpha = 0.8$ ($4.19 \times 10^{-5}$) is consistent with this order of magnitude. Az-Zo'bi et al. [5] do not report MSE directly, but their qualitative observation that MABC-PINN requires more training epochs than a classical baseline is consistent with the pattern observed in the present work. A fully controlled comparison against published fPINN implementations is a natural direction for future work.

**Other nonsingular kernels and relation to Caputo–Fabrizio.** The ABC derivative is not the only fractional operator with a nonsingular kernel. The Caputo–Fabrizio (CF) operator [13] replaces the Mittag-Leffler kernel with an exponential kernel $e^{-\frac{\alpha}{1-\alpha}(t-\tau)}$, which is also bounded, nonsingular, and integrable on $[0, T]$. We do not include a CF discretization in the present comparison, for two reasons. First, the exponential kernel decays far more rapidly than the Mittag-Leffler kernel: for practical values of $\alpha$, the effective memory horizon of the CF kernel is a small fraction of the Mittag-Leffler horizon at the same $\alpha$, so the CF derivative behaves as a *short-memory* operator rather than the genuinely long-memory ABC operator. A CF-based PINN residual would therefore fall closer to the local-surrogate end of the diagnostic spectrum, adding a third but qualitatively similar case rather than a new class of behavior. Second, the CF operator is known not to satisfy some of the fundamental identities expected of a fractional derivative (e.g., the index law), and its use in numerical applications has been questioned [7]. A systematic extension of the diagnostic protocol to other nonsingular kernels, including tempered and generalized Mittag-Leffler kernels, is a natural direction for future work.

### 7.2 Limitations

The study is confined to one spatial dimension, one fractional order $\alpha = 0.8$, and a single PDE. The reference solutions are computed on grids that are sufficient for stability but not extrapolated to the continuous limit. The PINN training uses a fixed number of epochs; a study with matched wall-clock budget rather than matched epoch count is left for future work. Finally, no uncertainty quantification is incorporated. Extensions to additional fractional orders (beyond the two values $\alpha = 0.8$ and $\alpha = 0.5$ studied here), two-dimensional domains, other nonsingular kernels (e.g., Caputo–Fabrizio), and network-structured domains [20] are left for future work. The reported $\alpha$-dependence is based on three random seeds per configuration; larger seed ensembles would be required to establish statistical significance for the small differences observed at $\alpha = 0.5$ (naive FD versus Caputo-L1). We also note three further limitations. First, the reference solutions are verified for grid convergence via coarse–fine comparisons, but they are not cross-checked against an analytic manufactured solution; such a test would strengthen the confidence in the reference solvers. Second, the ABC-L1 discretization used here retains the full Mittag-Leffler kernel through quadrature, but no formal consistency or convergence analysis of the quadrature scheme is provided; such an analysis is left for future work. Third, the MSE and training-time ratios reported throughout compare each residual to its own reference solution and to the naive-FD baseline on the same architecture; because the three residuals solve different PDEs, these ratios quantify relative difficulty rather than a head-to-head accuracy comparison on a common ground truth.

## 8 Conclusion

We have introduced a three-test diagnostic protocol for detecting false fractionalization in physics-informed neural networks. Applied to the Atangana–Baleanu–Caputo derivative in a one-dimensional advection-dispersion equation, the protocol classifies the naive two-point FD residual as a local surrogate, the Caputo-L1 residual as genuine power-law fractional, and the ABC-L1 residual with Mittag-Leffler kernel as genuine nonsingular fractional. A controlled three-way comparison under identical architecture, optimizer, and epoch budget—in which each residual is evaluated against its own reference solution—shows that the price of genuine non-locality is a factor of 2.0× (Caputo) to 13.8× (ABC) in MSE, at 10.8×–13.8× the training cost of the local baseline. We recommend that future work apply the diagnostic protocol before labeling any numerical scheme as "fractional".

## Code and Data Availability

All code, reference solutions, PINN checkpoints, and figure-generation scripts, together with the exact parameter values used in all experiments, are available at `https://github.com/mrmlili/aria-icnabs`. The repository is archived on Zenodo at DOI 10.5281/zenodo.23266644.

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